

Introduction

Hello, I'm **Jai, A Power BI and SQL Specialist.**

I prepared an in-depth analysis of the **E-Commerce Dataset** to identify meaningful business insights and improve decision-making through data analysis.

The main objective of this project is to analyse sales performance based on different factors such as:

- Month-wise Sales Analysis
- Day-wise and Year-wise Sales Trends
- City-wise Sales Performance
- Staff Performance Analysis
- Overall Sales Insights and KPI Tracking

Overview:

- ❖ This project is overall analysis of **sales performance ,trends ,KPI's and deep insights.**
- ❖ This project helps us to understand sales pattern, customer behaviour and prediction.
- ❖ Data driven insights used to take better strategic and business decisions .
- ❖ Overall project is developed graphically in Power BI.

Data transformation:

(A) Data cleaning:

there are total 9 csv files which is cleaned properly by following these steps:

1. Remove blank rows and duplicate records.
2. remove extra spaces in the datasets.
- 3.detect and correct datatypes of columns specially in date column in 'yyyy-mm-ddd' format.
- 4.handed missing values.

(B) Data Modeling:

1. import all csv files into SQL one by one.

- 2.there are total 9 csv files which is imported detecting correct data types:

(Brands , categories , customers , order_items , orders , products , staff , stocks , stores.)

3. mapping cardinality between tables in the form of different types of relationship.

- One to one relationship.
- Many to one relationship.
- Many to many relationship.
- One to many relationship.

4. making relationship between tables is to extracting required kpi's ,sales ,and trends.

(c) performed sql queries for get sales informations of staff wise,customer wise,state wise following :

Join,group by,orderby clause,creating vies for reusable insights.

Project visualizations:

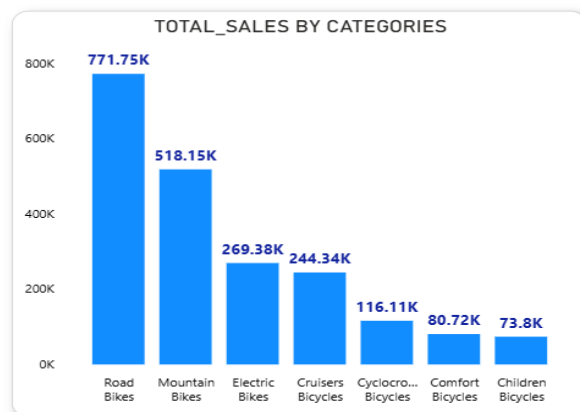
(1) kpi's:

- Total Sales 2.07M
- Total Orders 387
- Total customers 1K
- Average Orders 5.36K

(2) Charts Visualizations:

(A) Total Sales by category.

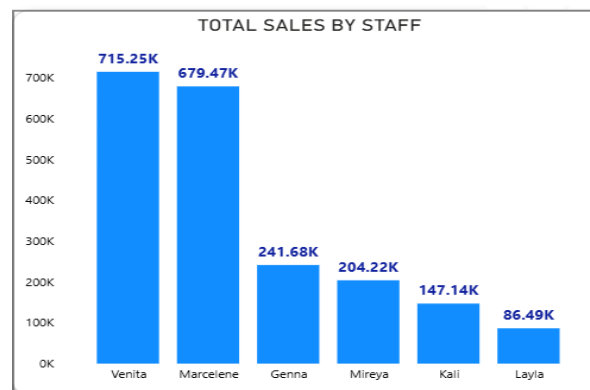
Roa Bikes is the highest category in sales: 771.75k.



(B) Total sales by staff:

Venita Staff is goes to hights sales 715.25K.

Layia staff perform not well 86.49 lowest sales.



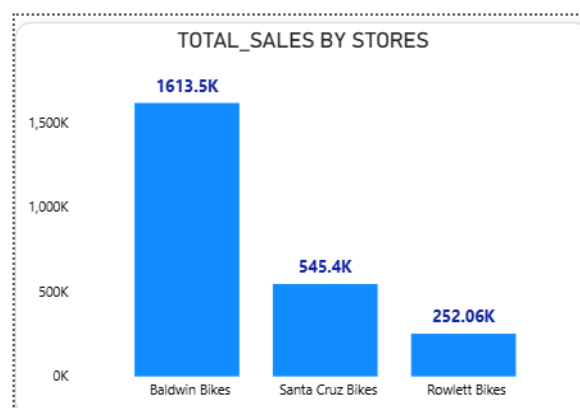
(c) total sales by stores:

Baldwin bikes store hits highest sales 1613.5k

and Rowlett Bikes stores is shows

Lowest saes 252.06k.

(d) Regional Sales Distribution:

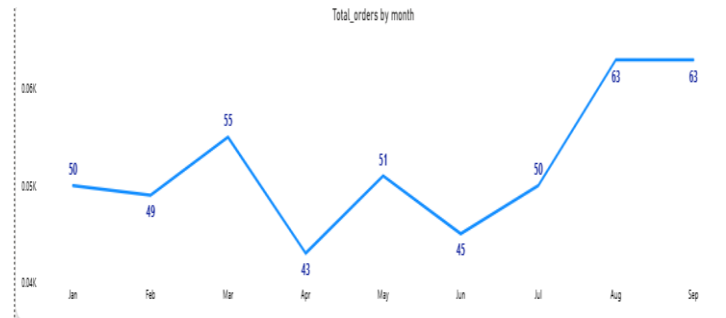


(E) Line charts :

(A) Month wise orders:

In august and September month orders were very high 62 orders came.

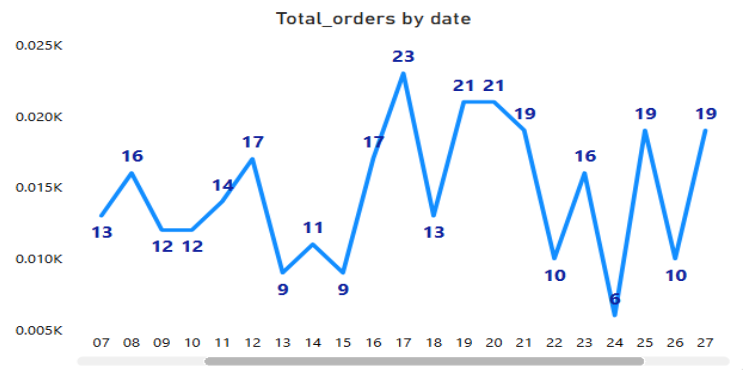
In April month orders came 43 which is very low.



(B) date wise orders:

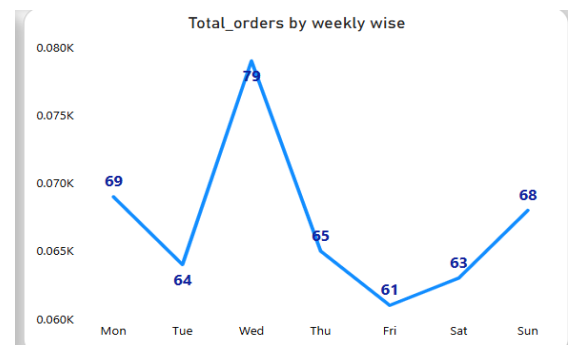
On 17th day highest orders came 23

On 24th day lowest orders came 6



(c) weekly orders:

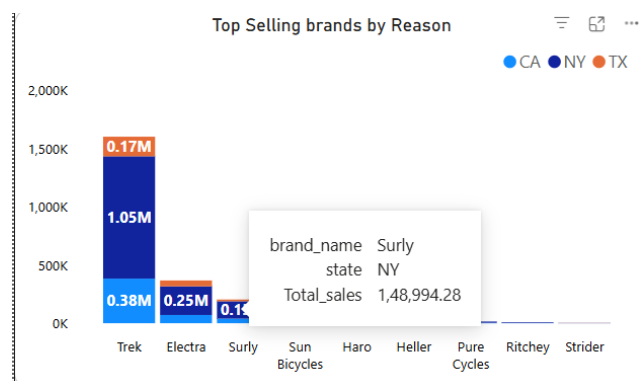
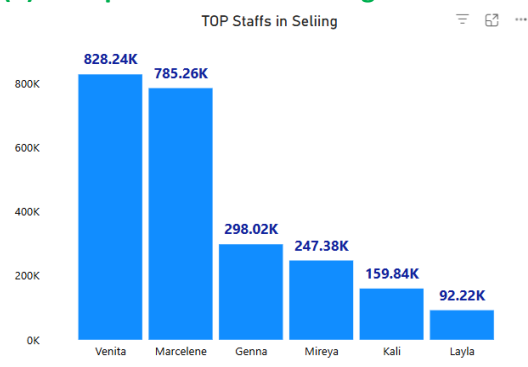
On Wednesday 79 orders came.
and lowest orders came 61



(3) Busines insights:

(1) Top selling brand is **Trek** where highest selling did **NY State** approx **1.05 M**.

(2) Staff performance in selling:



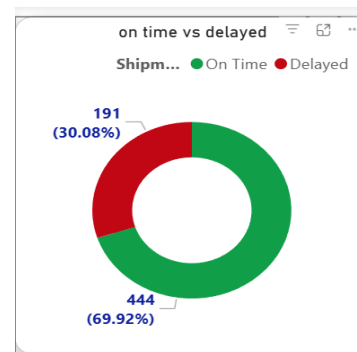
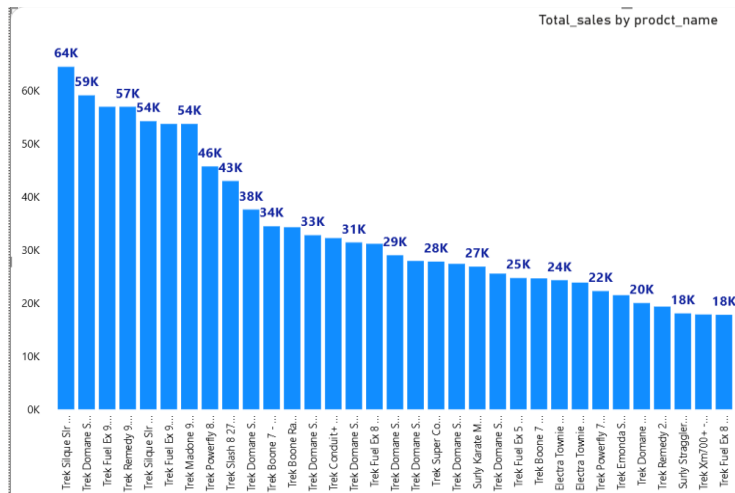
Topp staff name is Venita which sales is incredible around **828.24k**
and lowest sales saff is Layla.**92.22k**

(3) customer orders and fulfilment status.

mostly customer satisfied because on time delivery.

70% but 30% are not satisfied.

(4) most profitable product categories:



Trek slique slr 7 women's -2017 product is goes to **64k** In top sales.

Second product category is Trek Domane slr 6 disc-2017

Recommendations:

- **Focus on High-Performing Categories and Brands**
 - Increase inventory, marketing campaigns, and promotional offers for **Road Bikes** and **Trek products**, as they generate the highest sales revenue.
 - **Business Impact:** Higher product availability and visibility can increase revenue and market share.
- **Improve Low-Performing Staff Performance**
 - Conduct sales training, mentoring, and performance reviews for lower-performing staff members while replicating the sales strategies used by top performers such as Venita.
 - **Business Impact:** Improved employee productivity can increase overall sales and customer engagement.
- **Strengthen Sales in Low-Performing Stores**

- Analyze the reasons behind lower sales at stores such as **Rowlett Bikes** and implement targeted promotions, local marketing campaigns, and inventory optimization.
- **Business Impact:** Better store performance can boost total company revenue and improve regional market penetration.
- **Enhance Customer Satisfaction and Order Fulfillment**
 - Address the 30% customer dissatisfaction rate by improving delivery timelines, order tracking, and customer support processes.
 - **Business Impact:** Higher customer retention, increased repeat purchases, and stronger brand loyalty leading to long-term revenue growth.